

# Sensor Technologies for Wastewater Collection Networks: Characteristics, Applications, and Limitations

## Abstract

Wastewater collection networks can be monitored for water level, flow, water quality, gas conditions, and pipe condition. Each purpose requires a different sensor or inspection method. This review compares the main technologies used in sewer networks, with emphasis on their practical use, strengths, and limitations. Non-contact radar and ultrasonic sensors are widely applicable to level monitoring, although their performance depends on installation and environmental conditions. Continuous level data can also support site-specific early detection of developing sewer chokes when current measurements are compared with a local historical pattern. Recent placement research shows that sensor reliability, consequence, elevation, information redundancy, and maintenance constraints must also be considered when selecting a small sensor set. Flow estimates require both hydraulic context and suitable calibration. Water-quality probes, UV/Vis instruments, and fluorescence sensors can identify changes in wastewater composition, but fouling and matrix effects remain important. Hydrogen sulfide monitoring also requires a clear distinction between gas-phase and liquid-phase measurements. The review shows that sensor selection should begin with the operational question and the conditions at the intended site.

**Keywords:** wastewater collection; sewer monitoring; level sensing; flow sensing; water quality; optical spectroscopy; hydrogen sulfide; operational monitoring.

## 1 Introduction

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Sewer monitoring is used to detect surcharge and overflow risk, estimate hydraulic loading, identify unusual discharges, manage odour and corrosion, and assess pipe condition. These tasks do not produce the same type of information. A level sensor records hydraulic behaviour, a water-quality probe responds to changes in wastewater composition, and an inspection system examines the physical condition of the pipe. Telemetry then carries the measurements from the sewer to an operator or data platform.

The useful characteristics of a sensor depend on its location and purpose. Humidity, condensation, foam, solids, corrosion, limited access, radio coverage, and power availability can all influence performance. The following sections group sensor technologies by the practical question they answer and compare their applications and limitations.

## 2 Monitoring Questions and Sensor Categories

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The literature is clearer when grouped by the question being answered rather than by a single list of hardware names.

Table 1: Question-led classification of sewer monitoring technologies.

| Question                         | Examples                                                                            | Information obtained                                                                                | Main limitation                                                                         | Sources (Ref., pp.)                                                                                                          |
|----------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| How full is the sewer?           | Radar, ultrasonic, short-range LiDAR/ToF, float or tilt switches                    | Continuous level trends or discrete surcharge events                                                | Humidity, condensation, foam, obstructions, fouling, or loss of trend information       | [6, pp. 3–4, 28–29]; [5, pp. 1–3, 6–9]; [24, pp. 7, 10]; [23, pp. 3–1–3–4]                                                   |
| How much is flowing?             | Contactless ultrasonic flow, level plus geometry, pump drawdown or hydraulic models | Flow or velocity estimates and hydraulic loading                                                    | Installation geometry, flow direction, backwater, and calibration assumptions           | [1, pp. 1–4, 11–18]; [6, pp. 13, 23–24]                                                                                      |
| What is in the wastewater?       | pH, conductivity, temperature, turbidity, UV/Vis, fluorescence, H <sub>2</sub> S    | Chemical changes, illicit-discharge indicators, organic-load proxies, odour or corrosion indicators | Fouling, matrix effects, local calibration, and phase-specific measurement              | [19, pp. 2–4, 11–12]; [12, pp. 1, 3–4, 8–9]; [11, pp. 1–2, 10–11]; [3, pp. 1, 5–6, 12]; [15, pp. 1–5, 10–15]; [4, pp. 13–16] |
| Is the asset blocked or damaged? | Acoustic screening, CCTV, computer vision, fibre-optic or other condition methods   | Structural or blockage information                                                                  | Usually an inspection or diagnostic workflow rather than a continuous level time series | [2, pp. 1–5]; [27, pp. 1–8]; [18, pp. 1–4, 9–12]; [14, pp. 1–6]; [17, pp. 1–6]                                               |
| How does the reading get back?   | LoRaWAN, cellular IoT, gateways, data platforms, soft sensors                       | Remote transmission and estimation of unmeasured states                                             | Coverage, power, data quality, model dependence, and site-specific validation           | [6, pp. 15–19]; [20, pp. 1–6, 12–15]; [13, pp. 1–7, 13–18]                                                                   |

Telemetry and soft sensors are enabling layers rather than physical hydraulic or water-quality sensors. Acoustic and CCTV methods are also distinct from fixed probes because they examine asset condition instead of producing a continuous level or chemical measurement.

Sensor technology is only one part of the placement decision. A small layout should be judged by the response it enables: early warning, I&I detection, flow reconstruction, source identification, condition inspection, or operational control. The placement literature also warns that methods developed for pressurised water-distribution networks cannot be transferred directly to gravity sewers without changing the hydraulic representation and validation criteria [25, 26].

### 3 Level and Depth Sensing

Continuous level data provide a time series of depth from which a utility may examine normal patterns, surcharge, and rate of change. A binary float or tilt switch provides an event threshold but cannot produce the same trend information. The choice therefore depends on whether the monitoring purpose is broad event notification, hydraulic diagnosis, or both.

#### 3.1 Non-contact Radar

Radar level sensors use an electromagnetic signal reflected from the wastewater surface to measure the distance between the sensor and the water. Because the sensor is mounted above the flow, it avoids direct contact with wastewater. Its performance still depends on the selected instrument, mounting position, beam geometry, and site conditions.

Drenoyanis et al. [6] used non-contact radar for sewer level measurement and pumping-station drawdown tests. The study describes a portable drawdown device and a LoRaWAN-enabled prototype installed in the Shoalhaven Water network. A separate radio survey recorded communication distances of up to 20 km in an urban area and up to 35 km under line-of-sight conditions. These distances describe the LoRaWAN link, not the radar measurement range.

Radar was less affected by humidity and mist than ultrasonic sensing in the comparison reported by Drenoyanis et al. Steam, condensation, foam, and changes in air temperature were identified as potential sources of unstable ultrasonic readings. Radar nevertheless requires careful installation because an unsuitable mounting arrangement can produce false detections. The study also identifies its higher cost as a disadvantage [6].

Drenoyanis et al. [6, pp. 28–29] describe an integrated 80 GHz short-range radar network with approximately five devices around each pumping station as a future development. Instrument frequency, accuracy, service life, and power requirements therefore need to be considered for the particular radar unit selected.

### 3.2 Ultrasonic, LiDAR, and Time-of-Flight Sensors

An ultrasonic level sensor estimates distance from the travel time of an acoustic pulse reflected by the wastewater surface. It is a common non-contact option and can provide a continuous level record at relatively low power. Steam, mist, condensation, foam, and temperature-dependent changes in sound speed can cause unstable readings [6, pp. 3–4]. Installation must also provide a clear acoustic path between the sensor and the wastewater surface.

Do et al. [5] used continuous ultrasonic level data to detect developing sewer chokes in Stonyfell, South Australia. Their case study examined 62 datasets collected over one year and showed how changes in the normal level pattern can provide an earlier indication of a blockage.

Utepov et al. [24] compared Sharp infrared, TF-Luna LiDAR, TOF400 laser, and JSN-SR04T ultrasonic sensors in a prototype designed to imitate a sewer well. In that arrangement, the JSN-SR04T measured to approximately  $\pm 5$  cm at distances up to 2 m, while the TF-Luna result was within approximately  $\pm 8$  cm at distances up to 5 m. The reported material cost of the complete prototype was about USD 170.07, excluding labour. The tests were conducted in an imitative setup, and the authors identified trials in different sewer systems as future work.

LiDAR and other optical time-of-flight sensors offer compact distance measurement and may suit locations with a clear optical path. Their exposed optical surfaces can be affected by condensation and deposits. Field use therefore depends on protection of the sensing window, suitable mounting geometry, and testing under the conditions found in the target sewer.

### 3.3 Binary Float and Tilt Sentinels

Float and tilt devices answer a simpler question: has wastewater crossed a set threshold? Their switch output is useful for surcharge or overflow alerts when a continuous level record is unnecessary. A single switch cannot show the rate of rise or describe changes between its trigger and reset levels. US EPA guidance describes mechanical floats as depth sensors and notes that grease and solids can obstruct their movement in sewer applications [23]. Battery life, maintenance intervals, and cost depend on the selected switch, logger, and communication system.

## 4 Early Detection of Sewer Chokes and Overflows

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Continuous level sensing becomes operationally useful when the time series is interpreted against the normal behaviour of the monitored location. A fixed threshold can identify an absolute high level, but it may either miss a developing partial blockage or create too many alarms to support action. Do et al. [5, pp. 3–4] provide a South Australian example of this problem. Their Stonyfell pilot initially deployed 82 ultrasonic water-level sensors, of which 62 remained operational for approximately one year. The monitored network contained about 11,300 m of pipe and the deployed system covered an estimated 48% of the network. Fixed low- and high-priority alarms were set at 150 and 300 mm. The low-priority threshold generated more than 100,000 alarms in 2019 without operational action, motivating a pattern-based early-warning method.

### 4.1 From level thresholds to local patterns

The Stonyfell study identified two features associated with developing chokes: irregularly high level peaks and periods in which the level remained above its historical pattern for longer than usual [5, pp. 6–8]. The second feature is particularly useful for early warning because a partial obstruction can reduce hydraulic capacity before the level reaches a fixed high threshold. During peak inflow, water may surcharge upstream of the obstruction and remain elevated; when inflow falls, the level may temporarily return towards normal. Repeated holding and release periods can therefore provide evidence of a developing obstruction, although a sudden choke may not produce a useful warning period.

### 4.2 Finding irregular high-level durations

Do et al. formulated the Finding Irregular high-water-level Durations (FID) method as a site-specific statistical detector. For each incoming level measurement, the method compares the current value with the 95th-percentile level calculated from a two-hour window centred on the same time over the previous 14 days. A high-level period begins when the current value exceeds this local reference and ends when it falls below it. In the Stonyfell configuration, a low-priority alarm required the irregular period to last longer than 20 minutes, corresponding to four consecutive five-minute readings, with at least one reading above 50 mm, one-third of the typical 150 mm pipe diameter [5, pp. 8–10]. The alarm was intended to trigger data checking and field verification before a maintenance order, not to prove that a blockage was present automatically.

### 4.3 Reported performance and transfer limits

The method generated an alarm before all 24 confirmed choke or overflow events evaluated in the first analysis. The first warning occurred 8–57 days before the event, with an average lead time of 32.6 days [5, pp. 10–12]. In a separate comparison involving 41 choke events and 54 non-choke events, FID identified 40 of 41 choke events and produced 17 false-positive alarms, while CUSUM identified all 41 choke events but produced 42 false-positive alarms [5, pp. 12–14]. The comparison illustrates the operational trade-off between detection rate and alarm burden.

These results are evidence for a monitoring and information-gap strategy, not a universal performance guarantee. The FID parameters were selected for the Stonyfell network and should be recalibrated for a different pipe geometry, flow regime, sensor configuration, or event definition. Detection also depends on continuous data, sensor availability, time alignment, data cleaning, network coverage, and a practical response process. For sensor-placement studies, early warning should therefore be treated as a separate monitoring objective with explicit measures for event capture, lead time, false alarms, data continuity, and follow-up verification. Continuous ultrasonic or radar level sensing can provide the required time series, but the sensor location and the anomaly method must be selected jointly for the intended operational question.

## 5 Flow and Velocity Measurement

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For a partially full sewer, discharge depends on both the wetted cross-sectional area and the mean water velocity. A level sensor alone does not automatically provide a reliable flow measurement when the flow is non-uniform or the geometry and boundary conditions are uncertain.

### 5.1 Contactless Ultrasonic Flow

Alshami et al. [1] studied twelve sewer locations in Hong Kong using contactless ultrasonic flow and level sensors. The level–flow correlation coefficient was 0.36, and the average relative flow-rate error was 72.14% when the sensor results were compared with Manning-equation estimates. The result reflects the instruments, sites, and comparison method used in that study.

Possible causes of error included small pipe diameters, installation angle, inconsistent flow direction, water level, and the presence of multiple inlets or outlets. In practice, contactless flow sensors need site-specific installation and calibration, making flow measurement more demanding than level measurement alone.

### 5.2 Level-derived Flow and Manning Models

Manning-type level-to-flow estimates can be useful when their assumptions match the target sewer reach. In a blocked or backwatered reach, the water level may rise without a proportional increase in velocity or discharge. Reliable conversion therefore depends on hydraulic context, pipe geometry, calibration, and comparison with observed flow where possible. The 72.14% result reported by Alshami et al. illustrates this problem for one network and instrument configuration.

Drenoyanis et al. [6] used radar level measurements during pumping-station drawdown to estimate pump flow indirectly. This application differs from measuring flow in a partially full gravity sewer because it relies on the changing water volume within a defined wet well.

## 6 Water-Quality and Chemical Monitoring

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Hydraulic sensors show how full the sewer is and how its level or flow changes. Water-quality sensors provide different information: they can indicate an unusual discharge, dilution, changing organic load, or conditions associated with odour and corrosion. These sensors are most useful at locations where a chemical change has a clear operational meaning, such as an industrial inflow, a pressure-sewer discharge, or an upstream and downstream comparison point.

### 6.1 Physicochemical Event Detection

Shi et al. [19] deployed 22 low-cost units measuring level, temperature, and electrical conductivity across an urban Melbourne catchment for more than one year. Changes in these measurements were used to identify dry-weather illicit discharges and short-duration events. Electrical conductivity and temperature are useful event indicators, although the cause of a change still has to be interpreted from its timing, location, and surrounding conditions.

Li et al. [12] evaluated a sewage-quality monitoring system measuring temperature, conductivity, turbidity, and pH. The design included wall-jet cleaning and self-diagnostic functions. It was evaluated within a three-year field programme, and one trial site operated continuously in raw sewage for up to 24 months without maintenance or failure. This performance was achieved by the particular system and cleaning arrangement used in the study.

## 6.2 UV/Vis and Fluorescence

Lepot et al. [11] reviewed UV/Vis methods for estimating total suspended solids and total or dissolved chemical oxygen demand. These instruments provide rapid, reagent-free measurements, but the relationship between the optical signal and the target parameter changes with the wastewater matrix. Local calibration is therefore important when an instrument is moved to a new site or used with a different wastewater composition.

Carstea et al. [3] found that fluorescence features such as tryptophan-like Peak T and humic-like Peak C can correlate with biochemical oxygen demand, chemical oxygen demand, or total organic carbon. The strength of these relationships varies with the sample matrix and measurement method. Overlapping fluorophores, inner-filter effects, turbidity, and fouling can also alter the signal. Fluorescence is therefore most useful as a rapid calibrated proxy rather than a direct measurement of these laboratory parameters.

Optical instruments can provide frequent measurements without chemical reagents, which makes them useful for detecting short events between manual samples. Their main practical challenges are fouling, cleaning, calibration drift, and sensitivity to the local wastewater matrix. Cleaning may be provided by a wiper, flushing system, compressed air, or scheduled manual maintenance, depending on the instrument and installation.

Nutrient sensors include ion-selective and other electrochemical methods for ammonium, nitrate, and nitrite. Selectivity, interfering ions, temperature, drift, and membrane condition affect their use in complex water matrices [22]. Dissolved oxygen can be measured with amperometric or optical instruments; the two principles differ in response time, stability, flow dependence, and maintenance needs [9]. Microbial fuel-cell biosensors have also been investigated for continuous biochemical oxygen demand and toxicity monitoring, although biofilm behaviour and operating conditions influence their response [21].

## 7 Hydrogen Sulfide and Corrosion Monitoring

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Hydrogen sulfide is relevant to sewer odour, worker safety, and biogenic corrosion. Gas-phase sensors measure concentrations in the sewer headspace, while liquid-phase sensors measure dissolved sulfide in wastewater. The two measurements have different purposes, instruments, calibration needs, and suitable installation points.

### 7.1 Gas-Phase Sensors

Pacheco Fernández et al. [15] compared three commercial gas sensors at two sewer sites. OdaLog and MyData devices had stated ranges of 0–200 ppm, while the SulfiLogger range was 0–1000 ppm. The reported battery specifications were approximately eight months, three months at the stated transmission configuration, and two years for the respective devices. Measurement range, logging interval, and telemetry configuration therefore need to be considered together when selecting a gas sensor.

Agreement between the gas sensors was better at low concentrations, while deviations increased as the concentration rose. High humidity and device-management requirements were also important practical considerations. A suitable gas sensor therefore needs an adequate measurement range, resistance to sewer humidity, accessible calibration, and a logging interval that captures short concentration peaks.

### 7.2 Liquid-Phase Sensors

Despot et al. [4] studied online liquid-phase H<sub>2</sub>S sensors, including UV/Vis and electrochemical instruments. Local calibration, interference from solids, and regular inspection affected their operation. Useful monitoring points included the end of a pressure sewer and locations downstream of chemical dosing, where dissolved-sulfide measurements can help assess septicity and treatment performance.

Pandey et al. [16] reviewed sensor-based approaches to H<sub>2</sub>S monitoring and their role in understanding odour and corrosion. Long-term interpretation requires attention to sensor drift, the position of the sensor, ventilation, wastewater conditions, and the relationship between short concentration peaks and cumulative corrosion exposure.

## 8 Pipe Condition and Inspection Methods

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Acoustic screening, CCTV, computer vision, and fibre-optic methods examine structural defects, obstructions, and changes in pipe condition [2,14,17,18,27]. Acoustic methods can screen long pipe sections and help prioritise further inspection. CCTV provides direct visual evidence but requires access, suitable flow conditions, and skilled interpretation. Computer vision can assist with consistent defect classification, although its performance depends on image quality and the data used to train the model. These methods complement continuous hydraulic monitoring rather than replacing it.

## 9 Fouling, Maintenance, Telemetry, and Power

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Fouling, solids, condensation, calibration, and safe access influence the reliability and cost of sewer monitoring. Li et al. [12] demonstrated wall-jet cleaning for a multi-parameter water-quality system, while Despot et al. [4] found that liquid-phase  $H_2S$  sensors needed regular inspection and maintenance. Maintenance frequency depends on the sensor surface, wastewater characteristics, cleaning system, and ease of access.

Non-contact measurement avoids direct immersion and usually reduces fouling, but exposed surfaces and mounting hardware may still need inspection. Radar is sensitive to mounting and beam geometry, while ultrasonic and optical sensors can be affected by condensation or deposits. Contacting and submerged devices face greater exposure to solids, grease, and biofilm.

Drenoyanis et al. [6] provide an Australian LoRaWAN case study and regional signal survey. The recorded communication distances applied to the antenna arrangements and terrain used in that survey, so radio performance still needs to be tested at the intended monitoring sites. Sikorski et al. [20] show that sensor coverage, battery capacity, sampling frequency, and cost are linked decisions. More frequent measurement and transmission improve temporal detail but increase energy use and data volume.

Operational planning should include installation access, cleaning, calibration, communications testing, battery replacement, and data-quality checks. These requirements can outweigh the purchase price when a sensor is difficult to reach or maintain.

## 10 Sensor Placement and Operational Integration

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Sensor placement determines which network states can be observed, how quickly an abnormality can be recognised, and whether the resulting measurement can support a practical response. The additional placement literature is useful here because it separates direct sewer evidence from methods transferred from other network types.

### 10.1 Wastewater-specific multi-criteria decisions

Yao et al. [26] distinguish pressurised, fully filled water-distribution systems from gravity-driven, partially filled wastewater collection systems with stronger temporal variability and higher solids exposure. The distinction affects the hydraulic model, the sensing assumptions, and the meaning of coverage or detection. Their review describes Utopian Point and Compromise Programming as transparent ways to select one implementable layout from a Pareto set while balancing cost, coverage, detection performance, and robustness. It also identifies sparse deployment, model uncertainty, imperfect sensor behaviour, and maintenance as placement uncertainties. For a sewer project, these methods should be used to explain a decision between candidate layouts, not to claim a universal sensor count without local costs, priorities, and field validation.

### 10.2 Deterministic placement for inflow and infiltration

Wu et al. [25] address inflow and infiltration (I&I) in an urban sewer network. Their deterministic two-stage method first groups nodes using spectral clustering and joint entropy, then uses topology and average I&I detection reliability to select sensor locations. In the reported case, joint entropy had a correlation above 0.95 with detection reliability, and the proposed method gave approximately 7–15% higher detection reliability than the comparison heuristic methods. This supports adding a deterministic entropy-based baseline beside greedy and stochastic optimisers. The finding concerns a particular sewer network and large I&I events; it does not establish performance for every sewer or replace incident labels, hydraulic validation, or field inspection.

### 10.3 Consequence, redundancy, and elevation

Guan et al. [8] show in a water-distribution network that equal importance assigned to every node can produce redundant sensor layouts. Their multi-objective White Whale Optimization model combines a leak-sensitivity

function with pressure-correlation and shortest-path constraints. The transferable lesson for sewer placement is to weight candidates by environmental, customer, flooding, or maintenance consequence and to avoid placing adjacent sensors that provide identical evidence unless deliberate redundancy is required. The pressure-based equations and results should not be copied directly into a gravity-sewer model.

Jain and Reddy [10] provide a second cross-domain example. Their water-distribution leak-localisation method combines spectral clustering, topology and sensitivity-based centrality, three-dimensional coordinates, and a multilayer perceptron. Three-dimensional grouping produced more spatially diverse and topographically informed layouts in their case. This supports testing elevation, grade, drop structures, and hydraulic separation as sewer candidate features, but the reported leak-localisation accuracy is not a wastewater performance result.

#### 10.4 From detection to operational response

Garofalo et al. [7] connect sensors to a decentralised real-time control system with electronically movable gates and use of in-network storage to reduce simulated combined-sewer overflow and flooding volumes. The study demonstrates why a placement design should record the action that follows an alert and the required telemetry latency. It is nevertheless a simulated urban-drainage control study in Cosenza, Italy. For the present review, automated gates or pump control should be treated as a future operational layer requiring actuators, fail-safe logic, hydraulic safeguards, and client authority. A continuous level sensor can support detection and dispatch without proving that an overflow will be prevented.

### 11 Comparison of Sensor Characteristics

Table 2: Characteristics and applications of sewer monitoring technologies.

| Technology                  | Main use                                       | Useful characteristics                                 | Limitations                                                         | Sources (Ref., pp.)                                        |
|-----------------------------|------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------|
| Radar level                 | Continuous level and pump draw-down            | Non-contact; relatively resistant to humidity and mist | Careful mounting and clear beam path; higher cost                   | [6, pp. 1, 3–4, 28–29]                                     |
| Ultrasonic level            | Continuous level and early surcharge detection | Non-contact; low-power options are available           | Condensation, foam, temperature, and mounting conditions            | [5, pp. 1–3, 6–9]; [24, pp. 7, 10]; [6, pp. 3–4]           |
| LiDAR/ToF                   | Short-range distance measurement               | Compact and potentially low power                      | Optical fouling, condensation, and limited sewer field experience   | [24, pp. 7, 10, 15]                                        |
| Float/tilt switch           | Threshold alarm                                | Simple event notification                              | No continuous trend; movement can be obstructed by grease or solids | [23, pp. 3-1-3-4]; [20, pp. 1–6]                           |
| Contactless ultrasonic flow | Flow and level estimation                      | Avoids submerged installation                          | Sensitive to geometry, flow direction, water level, and calibration | [1, pp. 1–4, 11–18]                                        |
| Physicochemical probes      | Conductivity, temperature, pH, and turbidity   | Detects changes in wastewater composition              | Fouling, calibration drift, and interpretation of events            | [19, pp. 2–4, 11–12]; [12, pp. 1, 3–4, 8–9]                |
| UV/Vis and fluorescence     | TSS and organic-load proxies                   | Rapid and reagent-free                                 | Local calibration, matrix effects, and optical fouling              | [11, pp. 1–2, 10–11]; [3, pp. 1, 5–6, 12]                  |
| H <sub>2</sub> S sensors    | Headspace gas or dissolved sulfide             | Supports odour, corrosion, and dosing control          | Phase-specific calibration, humidity, solids, and maintenance       | [15, pp. 1–5, 10–15]; [4, pp. 13–16]; [16, pp. 1–4]        |
| Telemetry and soft sensors  | Remote data transfer and estimated states      | Supports distributed monitoring and automated alerts   | Coverage, power use, data quality, and model transfer               | [6, pp. 15–19]; [20, pp. 1–6, 12–15]; [13, pp. 1–7, 13–18] |

### 12 Matching Sensors to Monitoring Tasks

No single sensor describes every aspect of a wastewater network. A float or tilt switch suits a simple high-level alarm, while radar or ultrasonic sensing is more useful when the timing, rate, and duration of surcharge are important. Flow sensors are needed where hydraulic loading must be quantified. Water-quality and H<sub>2</sub>S instruments are better suited to selected locations where a change can be linked to a discharge source, corrosion risk, or treatment action. Acoustic screening and CCTV remain appropriate for structural defects and blockages.

In each case, the value of the measurement has to be considered alongside access, power, telemetry, fouling, calibration, and maintenance.

For a placement study, the selected layout should also be evaluated against event recall, warning lead time, false-alarm burden, I&I detection reliability, data continuity, verification delay, and life-cycle cost. A geographically broad layout may still provide poor operational value if its readings are intermittent or its alarms cannot be investigated. Conversely, a smaller set of continuous level sensors may be preferable when it gives a reliable time series for anomaly detection and a clear response pathway.

## 13 Conclusion

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Sewer monitoring is best approached through the operational question that needs to be answered. Radar, ultrasonic, LiDAR, and float devices provide different levels of information about surcharge and water level. Radar is attractive in humid environments, while ultrasonic and optical sensors may offer lower-cost alternatives where condensation, foam, and fouling can be controlled. Flow measurement is more demanding because pipe geometry, flow direction, backwater, and calibration all influence the result.

Water-quality and optical sensors add information that hydraulic measurements cannot provide, particularly for unusual discharges, organic-load changes, odour, and corrosion. Their usefulness depends on cleaning, calibration, and the ability to interpret the signal at a particular site. Gas-phase and liquid-phase H<sub>2</sub>S monitoring should also be selected separately because they measure different parts of the sewer environment. The added placement evidence strengthens the case for purpose-specific, consequence-weighted layouts and for a deterministic I&I baseline, while also showing why results from pressurised water-distribution networks must be adapted before use in gravity sewers. For early choke and overflow warning, the Stonyfell evidence shows why a continuous level series and a site-specific anomaly method can be more informative than a single threshold, while still requiring local calibration and operational verification. Overall, the most useful sensor is the one that provides enough information for a defined operational response and can be maintained reliably at the intended location.

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